Ex 1.2:-

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Echelon forms:- / reduced row encholon form:-

in reduced enchelon form $\left[\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & \end{array}\right]$

↳ These all should be 1 and below it Numbers should be 0.

if $\left[\begin{array}{ccc|c} 1 & 3 & -1 & 9 \\ 5 & 6 & 7 & 8 \\ 0 & 0 & 0 & 0 \end{array}\right]$

In this last row is 0 so here it can be many solutions. there is free variable then build parametric eq.

Q). $x+y-z=-2$
$2x-y+z=5$
$-x+2y+2z=1$
$\rightarrow \left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 2 & -1 & 1 & 5 \\ -1 & 2 & 2 & 1 \end{array}\right]$

Ans:- For the Encholon form first make all 3 1. Apply different operations on rows to make 1 and all below it to be 0. $\rightarrow$

$\left[\begin{array}{ccc|c} 1 & 1 & -1 & -2 \\ 0 & -3 & 3 & 9 \\ 0 & 0 & 4 & 8 \end{array}\right] \xrightarrow[\frac{1}{4}R_3]{-\frac{1}{3}R_2} \begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \left[\begin{array}{ccc|c} 1 & 1 & -1 & 2 \\ 0 & 1 & -1 & -3 \\ 0 & 0 & 1 & 2 \end{array}\right] \end{array}$ Now just find values.

if more unknowns then eqs so non-trivial sol.

Q. $2x+y-z=1$
$3x+2y+z=10$
$2x-y+2z=6$

Use gussion elemination subsitution to get answers :.

Ans:- In guaissan elemination just have to make these 3 '0' to solve.

$\left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 3 & 2 & 1 & 10 \\ 2 & -1 & 2 & 6 \end{array}\right] \xrightarrow{R_1+R_3} \left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 3 & 2 & 1 & 10 \\ 4 & 0 & 1 & 7 \end{array}\right] \begin{array}{l} \rightarrow R_1+1 \\ \\ \rightarrow \cancel{R_1+R_3} \end{array} \left[\begin{array}{ccc|c} 2 & 1 & -1 & 1 \\ 3 & 2 & 1 & 10 \\ 6 & 1 & 0 & 8 \end{array}\right]$

$\left[\begin{array}{ccc|c} 3 & 2 & 0 & 2 \\ 3 & 2 & 1 & 10 \\ 4 & 0 & 1 & 7 \end{array}\right] \xrightarrow{R_2-R_1} \left[\begin{array}{ccc|c} 3 & 2 & 0 & 2 \\ 0 & 0 & 1 & 8 \\ 4 & 0 & 1 & 7 \end{array}\right]$

—x—x—x—x—x—x—x—x—x—x—x—x—x—x—x—x—x—

Ex# 1.2:- (1-26).

Q1) (a). ~~Reduced Row E form~~ both
(b). both
(c). both
(d). ~~row~~ " both form
(e). ~~row enchelon form~~ both
(f). ~~Neither~~ (both)
(g). row encholon form

Q2) (a) ~~REF~~ ~~Both~~
(b) ~~Both~~
(c) REF
(d) REF
(e) NOT
(f). Neither
(g). REF

Q3) (a) Pivot col → 1, 2, 3
Pivot Rows → 1, 2, 3
(b) Pivot Rows → 1, 2, 3
Pivot col → 1, 2, 3
(c) Pivot Rows → 1, 3, 4
Pivot cols → 1, 3, 4
(d) Pivot Rows → 1, 2
Pivot cols → 1, 2

Q4) (a) P.R → 1, 2, 3
P.C → 1, 2, 3
(b) P.R → 1, 2, 3
P.C → 1, 2, 3
(c) P.R → 1, 3, 4
P.C → 1, 3, 4
(d). P.R → 1, 2
P.C → 1, 3

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Q3) (a). ~~Stam~~ Pivot Row → Row 1, Row 2, Row 3

Pivot col → col 1, 2, 3

$$\begin{array}{c} \begin{array}{cccc} a & b & c & \end{array} \\ \left[\begin{array}{ccc|c} 1 & -3 & 4 & 7 \\ 0 & 1 & 2 & 2 \\ 0 & 0 & 1 & 5 \end{array}\right] \end{array}$$

$\boxed{c=5}$ $a-3b+4c=7$

$b+2c=2$

$\boxed{b=-8}$ $a=-37$

(b).

Ans:- Pivot Row → 1, 2, 3

col → 1, 2

$$\begin{array}{c} \begin{array}{ccccc} a & b & c & d & \end{array} \\ \left[\begin{array}{cccc|c} 1 & 0 & 8 & -5 & 6 \\ 0 & 1 & 4 & -9 & 3 \\ 0 & 0 & 1 & 1 & 2 \end{array}\right] \end{array} R_1-8R_3$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 8 & -5 & 6 \\ 0 & 1 & 4 & -9 & 3 \\ 1 & 0 & 0 & -13 & -10 \end{array}\right] \to R_1-2R_2$$

$$\begin{array}{c} \begin{array}{ccccc} a & b & c & d & \end{array} \\ \left[\begin{array}{cccc|c} 1 & 0 & 0 & 13 & 0 \\ 0 & 1 & 4 & -9 & 3 \\ 1 & 0 & 0 & -13 & -10 \end{array}\right] \end{array}$$

$a+13d=0$

$a-13d=-10$ $a=-13\left(\frac{5}{13}\right)$

$-26d=-10$ $a=-5$

$d=5/13$

(c).

Ans:- PR → 1, 2, 3, PC → 1, 4

(d)

Ans:- P.C

$$\left[\begin{array}{ccc|c} 1 & -3 & 7 & 1 \\ 0 & 1 & 4 & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \begin{array}{l} \to P.R \\ \to P.R. \\ \\ \end{array}$$

system is inconsistent due to this line.

Q4).

(a). $c=7, b=0, a=-3$

(b). I.R

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -7 & 8 \\ 0 & 1 & 0 & 3 & 2 \\ 0 & 0 & 1 & 1 & -5 \end{array}\right]$$

P.C

$c+d=-5,\ b+3d=2,\ a-7d=8$

(c) Same

(d). Inconsistent.

Q6).

Ans:-

$$\begin{array}{c} \begin{array}{cccc} x_1 & x_2 & x_3 & \end{array} \\ \left[\begin{array}{ccc|c} 2 & 2 & 2 & 0 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & 4 & -1 \end{array}\right] \end{array} \to R_3-\tfrac{4}{1}R_1$$

$$\left[\begin{array}{ccc|c} 2 & 2 & 2 & 0 \\ -2 & 5 & 2 & 1 \\ 0 & -7 & -4 & -1 \end{array}\right] \to R_1+R_2$$

$$\left[\begin{array}{ccc|c} 2 & 2 & 2 & 0 \\ 0 & 7 & 4 & 1 \\ 0 & -7 & -4 & -1 \end{array}\right] \to R_2+R_3$$

$$\left[\begin{array}{ccc|c} 2 & 2 & 2 & 0 \\ 0 & 7 & 4 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right] \to R_3/7$$

$$\left[\begin{array}{ccc|c} 2 & 2 & 2 & 0 \\ 0 & 1 & 4/7 & 1/7 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

$x_2+\frac{4}{7}x_3=\frac{1}{7}$, $x_1+x_2+x_3=0$

$\boxed{x_2=\frac{1}{7}-\frac{4}{7}x_3}$ $x_3=t$ etc.

$$\left[\begin{array}{ccc|c} 3 & 6 & -3 & -2 \\ 0 & -2 & 3 & 1 \\ 0 & 0 & 0 & 6 \end{array}\right]$$

$-2b+3c=1,\ 3a+6b-3c=-2$

Q7).

$$\left[\begin{array}{cccc|c} 1 & -1 & 2 & -1 & -1 \\ 2 & 1 & -2 & -2 & -2 \\ -1 & 2 & -4 & 1 & 1 \\ 3 & 6 & 0 & -3 & -3 \end{array}\right] \to R_1+R_2$$

Ans:-

$$\left[\begin{array}{cccc|c} 3 & 0 & 0 & -3 & -3 \\ 2 & 1 & -2 & -2 & -2 \\ -1 & 2 & -4 & 1 & 1 \\ 3 & 0 & 0 & -3 & -3 \end{array}\right] \to R_1+R_4$$

$$\left[\begin{array}{cccc|c} 3 & 0 & 0 & -3 & -3 \\ 2 & 1 & -2 & -2 & -2 \\ -1 & 2 & -4 & 1 & 1 \end{array}\right] \to R_1/3$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & -1 \\ 2 & 1 & -2 & -2 & -2 \\ -1 & 2 & -4 & 1 & 1 \end{array}\right] \to R_3-2R_2$$

$$\left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & -1 \\ -5 & 0 & 0 & 5 & 5 \\ -1 & 2 & -4 & 1 & 1 \end{array}\right] \to R_2-5R_3$$

$$\begin{array}{c} \begin{array}{ccccc} x & y & z & w & \end{array} \\ \left[\begin{array}{cccc|c} 1 & 0 & 0 & -1 & -1 \\ -5 & 0 & 0 & 5 & 5 \\ 0 & -10 & 20 & 0 & 0 \end{array}\right] \end{array}$$

$x-w=-1 \to x-w=1$

$-5x+5w=5$ $-x+w=1$

$\boxed{w=1+x}$

$x-(1+x)=1$

$-10y+20z=0$

$2z=y$

Q8).

Ans:-

$$\left[\begin{array}{ccc|c} 0 & -2 & 3 & 1 \\ 3 & 6 & -3 & -2 \\ 6 & 6 & 3 & 5 \end{array}\right] \to R_1R_2$$

$$\left[\begin{array}{ccc|c} 3 & 6 & -3 & -2 \\ 0 & -2 & 3 & 1 \\ 6 & 6 & 3 & 5 \end{array}\right] \to R_3-2R_1$$

$$\left[\begin{array}{ccc|c} 3 & 6 & -3 & -2 \\ 0 & -2 & 3 & 1 \\ 0 & -6 & 9 & 9 \end{array}\right] \to R_3-3R_2$$

$$\cancel{\left[\begin{array}{ccc|c} 3 & 6 & -3 & -2 \\ 0 & -2 & 3 & 1 \end{array}\right]}$$

Q9). $\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ -1 & -2 & 3 & 1 \\ 3 & -7 & 4 & 10 \end{array}\right] \to R_1+R_2$ (see Q11, Q12)

Ans:-

$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 3 & -7 & 4 & 10 \end{array}\right] \to R_3-3R_1$

$\left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & -1 & 5 & 9 \\ 0 & -10 & -2 & -14 \end{array}\right] \to R_3-10R_2$

$\left[\begin{array}{ccc|c} 2 & 1 & 2 & 8 \\ 0 & -1 & 5 & 4 \\ 0 & 0 & -52 & -104 \end{array}\right] \to R_1+R_2$

$\left[\begin{array}{ccc|c} 1 & 0 & 7 & 8 \\ 0 & -1 & 5 & 4 \\ 0 & 0 & -52 & -104 \end{array}\right] \begin{array}{l} \to 7R_2-5R_1 \\ \to \cancel{7R_2-5R_1} \\ \end{array}$

$\left[\begin{array}{ccc|c} 1 & 0 & 7 & 8 \\ 0 & & & \\ 0 & 0 & -52 & -104 \end{array}\right]$ $\left[\begin{array}{ccc|c} -5 & -7 & 0 & -22 \\ 0 & -1 & 5 & 4 \\ 0 & 0 & -52 & -104 \end{array}\right]$

→ divide and get $x_3=2, x_2=1, x_1=1$

Q13) (i) No trivial solution (m<n)

Q14) Yes it has trivial solution

Q15). $\begin{bmatrix} 2 & 1 & 3 \\ 1 & 2 & 0 \\ 0 & 1 & 1 \end{bmatrix} \begin{array}{l} \\ \to 2R_2-R_1 \\ \to R_4 \end{array}$

Ans:-

$\begin{bmatrix} 2 & 1 & 3 \\ 0 & 3 & -3 \\ 0 & 1 & 1 \end{bmatrix} \to R_3-3R_2$

$\begin{bmatrix} 2 & 1 & 3 \\ 0 & 3 & -3 \\ 0 & 0 & 10 \end{bmatrix} \begin{array}{l} \to 1/2 \\ \to 1/3 \\ \to 1/10 \end{array}$ $\left[\begin{array}{ccc|c} 1 & 1/2 & 3/2 & 0 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right]$

$x_1=0, x_2=0, x_3=0$

Q16). $\begin{bmatrix} 2 & -1 & -3 \\ -1 & 2 & -3 \\ 1 & 1 & 4 \end{bmatrix} \to R_2+R_3$

Ans:-

$\begin{bmatrix} 2 & -1 & -3 \\ -1 & 2 & -3 \\ 0 & 3 & 1 \end{bmatrix} \to R_2+R_1$

$\begin{bmatrix} 2 & -1 & -3 \\ 1 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix} \to R_1-2R_2$

$\begin{bmatrix} 2 & -1 & -3 \\ 0 & -3 & -3 \\ 0 & 3 & 1 \end{bmatrix} \to R_2+R_3$

$\begin{bmatrix} 2 & -1 & -3 \\ 0 & -3 & -3 \\ 0 & 0 & -2 \end{bmatrix}$

$\begin{bmatrix} 1 & -1/2 & -3/2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix}$

$x_3=0, x_2=x_1=0$

Q10). $\left[\begin{array}{ccc|c} 2 & 2 & 2 & 0 \\ -2 & 5 & 2 & 1 \\ 8 & 1 & 4 & -1 \end{array}\right] \to R_1+R_2$

Ans:-

$\left[\begin{array}{ccc|c} 2 & 2 & 2 & 0 \\ 0 & 7 & 4 & 1 \\ 8 & 1 & 4 & -1 \end{array}\right] \to R_3-R_2$

$\left[\begin{array}{ccc|c} 2 & 2 & 2 & 0 \\ 0 & 7 & 4 & 1 \\ 8 & -6 & 0 & -2 \end{array}\right] \to R_1/2$

$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 0 \\ 0 & 7 & 4 & 1 \\ 8 & -6 & 0 & -2 \end{array}\right] \to R_2-7R_1$

$\left[\begin{array}{ccc|c} -7 & 0 & -3 & 1 \\ 0 & 7 & 4 & 1 \\ 8 & -6 & 0 & -2 \end{array}\right] \to R_2 \to R_3$

$\left[\begin{array}{ccc|c} -7 & 0 & -3 & 1 \\ 8 & -6 & 0 & -2 \\ 0 & 7 & 4 & 1 \end{array}\right] \to R_1/-7$

$\left[\begin{array}{ccc|c} 1 & 0 & 3/7 & -1/7 \\ & & & \\ 0 & 7/4 & & \end{array}\right]$ ∴ See Q6 working.